SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

COURSE NAME: DEEP LEARNING **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks:25**

1. CNN vs Fully Connected Neural Network

Introduction

- CNN stands for Convolutional Neural Network.
- A Fully Connected Neural Network connects every neuron to the next layer.
- Both networks can be used for image classification.
- CNNs are specially designed to handle image and spatial data.

Architectural Design

CNN

- Contains convolutional layers, activation layers and pooling layers.
- Uses local connections between pixels and neurons.
- Preserves spatial information of images.
- Uses shared filters to extract features.

Fully Connected Neural Network

- Every neuron is connected to all neurons in the next layer.
- Images are usually flattened into a one-dimensional vector.
- Spatial relationships between nearby pixels are not directly preserved.
- Requires a large number of weights.

Spatial Feature Extraction

- CNN extracts features from small regions of an image.

- Convolutional filters detect edges, corners and textures.
- Deeper layers learn shapes and complex patterns.
- Pooling reduces image dimensions while retaining important features.

Parameters and Computational Complexity

- Fully connected networks require many parameters for large images.
- CNNs use small filters repeatedly across the image.
- The same filter weights are shared at different locations.
- This reduces memory usage and computational requirements.

Why CNN is Suitable for X-ray Images

- Preserves spatial relationships.
- Automatically extracts important features.
- Detects abnormalities and patterns.
- Requires fewer parameters.
- Provides better performance for image classification.

Conclusion

CNNs are more suitable for X-ray classification because they efficiently learn spatial features using convolution, filters and pooling while reducing the number of parameters.

2. Different CNN Layers and Filters

Introduction

CNNs contain different layers that work together to identify defects in industrial products.

Convolutional Layer

- Applies filters to the input image.
- Extracts important visual features.
- Detects edges, lines, textures and shapes.
- Produces feature maps.
- Deeper convolutional layers learn complex defect patterns.

Activation Layer

- Usually uses ReLU activation.
- Introduces non-linearity into the network.
- Helps the CNN learn complex relationships.
- Improves the ability to distinguish different defect patterns.

Pooling Layer

- Reduces the size of feature maps.
- Reduces computational requirements.
- Removes less important information.
- Max pooling selects the strongest feature from a region.
- Makes the network less sensitive to small changes in position.

Fully Connected Layer

- Usually appears near the end of the CNN.
- Combines extracted features.
- Performs final classification.
- Can classify products as normal or defective.

Early-Layer Filters

Early layers generally detect:

- Edges
- Lines
- Corners
- Basic textures
- Simple patterns

Deeper-Layer Filters

Deeper layers generally detect:

- Complex shapes
- Object parts
- Surface patterns
- Cracks
- Scratches
- Other defect-related features

Conclusion

All CNN layers work together to detect industrial defects. Convolution extracts features, activation adds non-linearity, pooling reduces dimensions and fully connected layers perform classification.

3. Parameter Sharing vs Independent Weights

Introduction

High-resolution surveillance images contain a large number of pixels. Efficient processing requires a network with fewer parameters and lower memory requirements.

Parameter Sharing in CNN

- CNNs use the same filter weights across different image locations.
- A filter scans the entire image.
- The same feature can be detected at different positions.
- This significantly reduces the number of parameters.

Independent Weights in Fully Connected Networks

- Each connection generally has its own weight.
- Large images create a huge number of connections.
- Memory requirements become very high.
- Training can become computationally expensive.

Comparison

CNN with Parameter Sharing	Fully Connected Network
Fewer parameters	More parameters
Lower memory usage	Higher memory usage
Efficient feature detection	Computationally expensive
Preserves spatial information	Spatial structure is less directly preserved
Same filter reused	Independent weights

Translation-Related Feature Detection

- A CNN can detect the same feature at different locations.
- For example, a surveillance system can detect an object whether it appears on the left or right side of an image.
- Pooling can provide additional tolerance to small position changes.

Importance in Surveillance

- Thousands of images can be processed more efficiently.
- Reduces memory requirements.
- Speeds up learning and inference.
- Allows detection of objects and suspicious patterns at different locations.

Conclusion

Parameter sharing makes CNNs highly efficient for large-scale image processing. Reusing filter weights reduces parameters while allowing important features to be detected throughout an image.

4. AlexNet vs ResNet

Introduction

- AlexNet and ResNet are important CNN architectures.
- AlexNet achieved major success in large-scale image classification.
- ResNet introduced residual connections for training very deep networks.

AlexNet

- Introduced in 2012.
- Contains 5 convolutional layers and 3 fully connected layers.
- Uses ReLU activation.
- Uses convolutional filters to learn image features.
- Has approximately 60 million parameters.
- Relatively easier to understand and train because it is shallower.

ResNet

- Introduced in 2015.
- Available in different depths such as ResNet-18, ResNet-34, ResNet-50 and ResNet-152.
- Uses residual blocks.
- Contains shortcut or skip connections.
- Supports training of very deep networks.
- ResNet-50 has approximately 25.6 million parameters.

Feature Extraction

AlexNet

- Early layers detect edges and textures.
- Middle layers learn shapes.
- Deeper layers learn object-level features.

ResNet

- Learns hierarchical features through many convolutional layers.
- Deeper networks can learn more complex representations.
- Skip connections improve information and gradient flow.

Vanishing Gradient Problem

- Very deep neural networks can suffer from vanishing gradients.
- This makes training difficult.
- ResNet uses skip connections to provide a direct path for gradients.
- This makes deep networks easier to train.

Computational Cost

- AlexNet has a large number of parameters, especially in fully connected layers.
- ResNet can be computationally expensive depending on its depth.
- However, modern ResNet models generally provide stronger feature extraction and better efficiency than AlexNet for many tasks.

Recommendation

ResNet is recommended for complex image recognition because:

- It supports deeper networks.

- It reduces degradation problems.
- Skip connections improve training.
- It learns complex features.
- It provides strong classification performance.

Conclusion

AlexNet was an important milestone in CNN development, while ResNet is generally more suitable for modern and complex image recognition applications.

5. Regularized CNN vs Non-Regularized CNN

Problem

- The CNN achieves very high training accuracy.
- It performs poorly on unseen images.
- This indicates **overfitting**.
- The model is memorizing training data instead of learning general patterns.

CNN Without Regularization

- High risk of overfitting.
- May memorize noise in training images.
- Very high training accuracy.
- Poor validation and testing performance.
- Less reliable for real-world applications.

Dropout

- Randomly disables some neurons during training.
- Prevents neurons from depending too much on each other.
- Reduces overfitting.
- Encourages the network to learn more robust features.
- Dropout is normally disabled during testing.

Data Augmentation

Data augmentation creates variations of training images.

Examples:

- Rotation
- Flipping
- Cropping
- Scaling
- Brightness changes
- Contrast changes

Benefits

- Increases training-data diversity.
- Helps the model handle different leaf orientations.
- Improves generalization.
- Reduces overfitting.

Batch Normalization

- Normalizes intermediate activations during training.
- Makes training more stable.
- Can improve optimization speed.
- Can provide a regularizing effect.
- Adds some computational operations during training.

Comparison

Non-Regularized CNN	Regularized CNN
High overfitting risk	Reduced overfitting
Poor generalization	Better generalization

Non-Regularized CNN	Regularized CNN
May memorize training data	Learns robust features
Simpler training	Slightly more complex training
Poor unseen-data performance	Better unseen-data performance

Recommended Approach

For plant disease classification:

1. **Data augmentation** should be used to increase image diversity.
2. **Batch normalization** should be used to improve training stability.
3. **Dropout** can be used when overfitting is still significant.
4. **Validation data** should be used to monitor performance.
5. **Early stopping** can prevent unnecessary training.

Conclusion

A regularized CNN is more reliable than a non-regularized CNN for plant disease classification. Combining data augmentation, batch normalization and suitable dropout can reduce overfitting and improve performance on unseen leaf images.

Code of Conduct:

I K Gowthami(192525428) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Gowthami

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided

Synthesis & Insight Generation	**20%**	**Generates meaningful insights and conclusions from comparisons**	**Some insights derived from comparison**	**Limited or unclear insights**	**No insights generated**
Clarity	**10%**	**Information is clearly presented (tables/figures) with logical structure**	**Generally clear with minor issues**	**Presentation lacks clarity or structure**	**Poor presentation and difficult to understand**